

Visibility Metrics for Java

References

- [Visibility Metrics and the Importance of Hiding Things](#)
- [Calculate metrics](#)
- [Controlling Access to Members of a Class](#)
- [Neo4j Python Driver](#)

Relative Visibility Of Types

A Java class or interface may be declared with the modifier `public`, in which case it is visible to all classes everywhere. If a class or interface has no modifier (the default, also known as package-private), it is visible only within its own package.

The relative visibility is the number of inner components that are visible outside (public) divided by the number of all types:

$$relativevisibility = \frac{public\ types}{all\ types}$$

Using package protected types is one of many ways to improve encapsulation and implementation detail hiding.

How to apply the results

The relative visibility is between zero (all types are package protected) and one (all types are public). A value lower than one means that there are types that are declared package protected. The lower the value is, the better implementation details are hidden.

Non public classes can't be accessed from another package so they can be changed without affecting code in other packages. They clearly indicate functionality that only belongs to one package. This also motivates to use more classes and to split up code into smaller pieces with a single responsibility and reason to change.

Table 1a - Top 40 artifacts with lowest median of package protection encapsulation

This table shows the relative visibility statistics aggregated for all packages per artifact and focusses on artifacts with many packages and hardly any package protected types (lowest median, high visibility). Package protected types would help to improve encapsulation.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Global_relative_visibility_statistics_for_types

	artifact	all	public	min	max	average	percentile25	percentile50	percentile75	percentile90
0	axon-common-5.0.1	150	123	0.615385	1.000000	0.876656	0.782609	1.000000	1.000000	1.000000
1	axon-spring-boot-autoconfigure-5.0.1	70	54	0.558824	1.000000	0.901261	0.875000	1.000000	1.000000	1.000000
2	axon-conversion-5.0.1	30	27	0.700000	1.000000	0.925000	0.925000	1.000000	1.000000	1.000000
3	axon-update-5.0.1	23	23	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
4	axon-tracing-opentelemetry-5.0.1	5	5	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000
5	axon-modelling-5.0.1	93	85	0.692308	1.000000	0.903907	0.867521	0.900000	1.000000	1.000000
6	axon-metrics-micrometer-5.0.1	13	10	0.750000	1.000000	0.875000	0.812500	0.875000	0.937500	0.975000
7	axon-messaging-5.0.1	570	432	0.250000	1.000000	0.808399	0.750000	0.846154	1.000000	1.000000
8	axon-test-5.0.1	73	59	0.600000	1.000000	0.819906	0.709677	0.833333	0.956522	0.982609
9	axon-server-connector-5.0.1	72	58	0.642857	0.892857	0.788495	0.727273	0.833333	0.846154	0.874176
10	axon-eventsourcing-5.0.1	100	66	0.200000	0.900000	0.599721	0.480769	0.666667	0.734921	0.813333

Table 1b - Top 40 artifacts with highest median of package protection encapsulation

This table shows the relative visibility statistics aggregated for all packages per artifact and focusses on artifacts with many packages and the highest median of package protected types (low visibility). Package protected types help to improve encapsulation.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Global relative visibility statistics for types

[illegible]

Table 1 Chart 1 - Relative visibility in artifacts

```
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_  
matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Para  
meters 'cmap' will be ignored  
    scatter = ax.scatter(  
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_  
matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Para  
meters 'cmap' will be ignored  
    scatter = ax.scatter(  
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_  
matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Para  
meters 'cmap' will be ignored  
    scatter = ax.scatter(  
<Figure size 640x480 with 0 Axes>
```

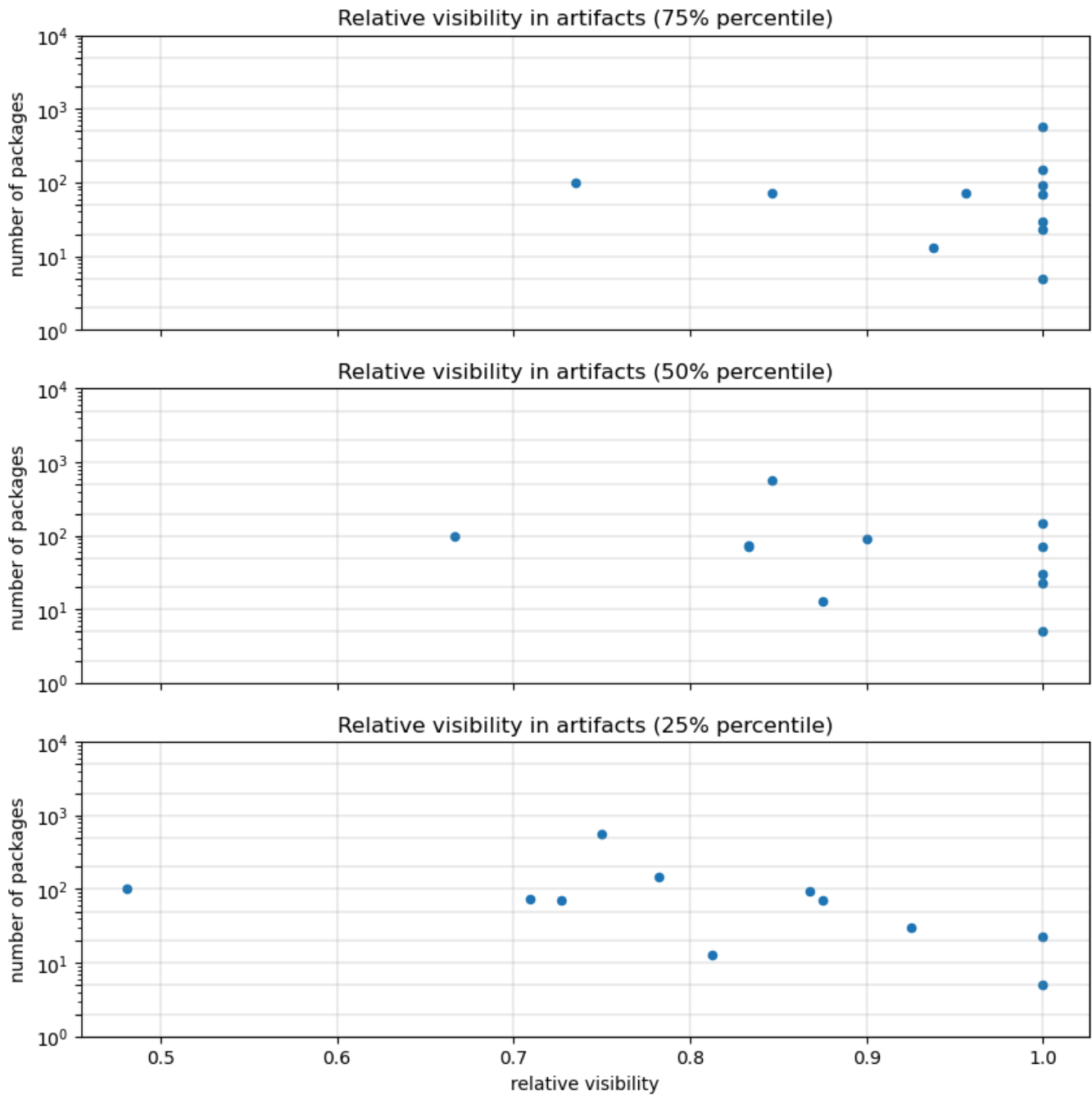


Table 2a - Top 40 packages with the highest visibility and lowest encapsulation

This table shows the relative visibility statistics per packages and artifact and focusses on packages with many types, hardly any package protected ones and therefore the highest relative visibility (lowest encapsulation). Package protected types would help to improve encapsulation.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Relative_visibility_public_types_to_all_types_per_package

	artifactName	fullQualifiedPackageName	packageName	publicTypes	allTypes	relativeVisibility
0	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot	springboot	21	21	1.0
1	axon-modelling-5.0.1	org.axonframework.modelling.entity.annotation	annotation	18	18	1.0
2	axon-modelling-5.0.1	org.axonframework.modelling	modelling	16	16	1.0
3	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conf...	configuration	14	14	1.0
4	axon-modelling-5.0.1	org.axonframework.modelling.entity.child	child	14	14	1.0
5	axon-common-5.0.1	org.axonframework.common.jdbc	jdbc	11	11	1.0
6	axon-conversion-5.0.1	org.axonframework.conversion.avro	avro	10	10	1.0
7	axon-common-5.0.1	org.axonframework.common.property	property	9	9	1.0
8	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	jdbc	9	9	1.0
9	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.sequ...	sequencing	9	9	1.0
10	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.tracing	tracing	7	7	1.0
11	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.tracing	tracing	7	7	1.0
12	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.util	util	7	7	1.0
13	axon-update-5.0.1	org.axonframework.update.configuration	configuration	7	7	1.0
14	axon-common-5.0.1	org.axonframework.common.util	util	6	6	1.0
15	axon-messaging-5.0.1	org.axonframework.messaging.tracing.attributes	attributes	6	6	1.0
16	axon-messaging-5.0.1	org.axonframework.messaging.core.correlation	correlation	6	6	1.0
17	axon-update-5.0.1	org.axonframework.update.api	api	6	6	1.0
18	axon-common-5.0.1	org.axonframework.common.lifecycle	lifecycle	5	5	1.0
19	axon-conversion-5.0.1	org.axonframework.conversion.converter	converter	5	5	1.0
20	axon-conversion-5.0.1	org.axonframework.conversion.json	json	5	5	1.0
21	axon-messaging-5.0.1	org.axonframework.messaging.monitoring.interce...	interception	5	5	1.0
22	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	store	5	5	1.0
23	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	subscribing	5	5	1.0
24	axon-tracing-opentelemetry-5.0.1	org.axonframework.extension.tracing.opentelemetry	opentelemetry	5	5	1.0
25	axon-update-5.0.1	org.axonframework.update	update	5	5	1.0
26	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	jpa	4	4	1.0
27	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.replay	replay	4	4	1.0
28	axon-update-5.0.1	org.axonframework.update.detection	detection	4	4	1.0
29	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	errorhandling	3	3	1.0
30	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	processing	3	3	1.0
31	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration...	reflection	3	3	1.0
32	axon-common-5.0.1	org.axonframework.common.jpa	jpa	2	2	1.0
33	axon-messaging-5.0.1	org.axonframework.messaging.core.configuration	configuration	2	2	1.0
34	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.conv...	conversion	2	2	1.0
35	axon-messaging-5.0.1	org.axonframework.messaging.core.conversion	conversion	2	2	1.0
36	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.gateway	gateway	2	2	1.0
37	axon-messaging-5.0.1	org.axonframework.messaging.core.reflection	reflection	2	2	1.0
38	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.actuato...	axonserver	2	2	1.0
39	axon-test-5.0.1	org.axonframework.test.server	server	2	2	1.0

Table 2b - Top 40 packages with the lowest visibility and highest encapsulation

This table shows the relative visibility statistics per packages and artifact and focusses on packages with many types, many package protected ones and therefore the lowest relative visibility (highest encapsulation). Package protected types help to improve encapsulation. Zero percent visibility and therefore packages with no public visible type are suspicious to be dead code.

Only the top 40 entries are shown. The whole table can be found in the following CSV report:

Relative_visibility_public_types_to_all_types_per_package

	artifactName	fullQualifiedPackageName	packageName	publicTypes	allTypes	relativeVisibility
0	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore.inm...	inmemory	1	5	0.200000
1	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	pooled	6	24	0.250000
2	axon-messaging-5.0.1	org.axonframework.messaging.monitoring.configu...	configuration	2	8	0.250000
3	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.inte...	interception	2	8	0.250000
4	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	inmemory	1	3	0.333333
5	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.in...	interception	2	5	0.400000
6	axon-messaging-5.0.1	org.axonframework.messaging.core.interception	interception	8	19	0.421053
7	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore.jpa	jpa	6	13	0.461538
8	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.configuration	configuration	7	14	0.500000
9	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	annotation	1	2	0.500000
10	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.an...	annotation	1	2	0.500000
11	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.autoconfig	autoconfig	19	34	0.558824
12	axon-messaging-5.0.1	org.axonframework.messaging.eventstreaming	eventstreaming	9	16	0.562500
13	axon-test-5.0.1	org.axonframework.test	test	3	5	0.600000
14	axon-common-5.0.1	org.axonframework.common.caching	caching	8	13	0.615385
15	axon-messaging-5.0.1	org.axonframework.messaging.core	core	50	80	0.625000
16	axon-common-5.0.1	org.axonframework.common.infra	infra	5	8	0.625000
17	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.event	event	9	14	0.642857
18	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.annotation.ref...	reflection	4	6	0.666667
19	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.inte...	interception	2	3	0.666667
20	axon-modelling-5.0.1	org.axonframework.modelling.configuration	configuration	9	13	0.692308
21	axon-conversion-5.0.1	org.axonframework.conversion	conversion	7	10	0.700000
22	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.repl...	annotation	7	10	0.700000
23	axon-test-5.0.1	org.axonframework.test.fixture	fixture	22	31	0.709677
24	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing	eventsourcing	5	7	0.714286
25	axon-messaging-5.0.1	org.axonframework.messaging.core.interception....	annotation	8	11	0.727273
26	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.dist...	distributed	8	11	0.727273
27	axon-server-connector-5.0.1	org.axonframework.axonserver.connector.query	query	8	11	0.727273
28	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.an...	annotation	9	12	0.750000
29	axon-metrics-micrometer-5.0.1	org.axonframework.extension.metrics.micrometer	micrometer	9	12	0.750000
30	axon-messaging-5.0.1	org.axonframework.messaging.core.timeout	timeout	6	8	0.750000
31	axon-messaging-5.0.1	org.axonframework.messaging.eventhandling.proc...	token	6	8	0.750000
32	axon-common-5.0.1	org.axonframework.common.annotation	annotation	3	4	0.750000
33	axon-messaging-5.0.1	org.axonframework.messaging.commandhandling.co...	configuration	3	4	0.750000
34	axon-messaging-5.0.1	org.axonframework.messaging.queryhandling.conf...	configuration	3	4	0.750000
35	axon-messaging-5.0.1	org.axonframework.messaging.core.unitofwork.tr...	transaction	3	4	0.750000
36	axon-spring-boot-autoconfigure-5.0.1	org.axonframework.extension.springboot.service...	connection	3	4	0.750000
37	axon-eventsourcing-5.0.1	org.axonframework.eventsourcing.eventstore	eventstore	34	45	0.755556
38	axon-messaging-5.0.1	org.axonframework.messaging.core.retry	retry	10	13	0.769231
39	axon-common-5.0.1	org.axonframework.common.configuration	configuration	36	46	0.782609

Table 2 Chart 1 - Relative visibility of packages

```
/home/runner/miniconda3/envs/codegraph/lib/python3.12/site-packages/pandas/plotting/_matplotlib/core.py:1351: UserWarning: No data for colormapping provided via 'c'. Parameters 'cmap' will be ignored
scatter = ax.scatter(
```

<Figure size 640x480 with 0 Axes>

